Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/06/2005
PI Name:	Iyoho, Anthony Ph.D.		
Project Title:	Modeling and Control of Bio-systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	65211	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	09/01/2003	**End Date:**	08/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Awardee says his end date was 8/31/2005 (previously listed as 8/31/2006)--3/29/07		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The first goal of this research was develop a more complete model of the human's thermoregulation to handle various applications involving astronauts in space, warfighters, fire fighters, combat pilots, hazardous waste workers, and other industries where comfort in thermally stressful environments needs to be insured. Specifically, the model includes: the arteriovenous anastomoses (AVAs), an accurate model of the human digits, and cold induced vasodilation (CIVD). Future work will be needed to 'individualize' the model to investigate the effect of physiological variations (gender, age, race, body composition, body type, etc.). In addtiion, the effect of microgravity on human thermal physiology will need to be quantified. The second part of this research was to develop a preliminary computational model for hysterical paralysis using neuro-circuitry information and systems biology principles. The model captures the salient features of the neuro-psychiatric disorder reported in the literature. Studies are on-going to model in more detail both the intra-cellular		

	and extra-cellular features of the model, within the framework proposed. The third part of this research was to develop a computational model of a pyramidal cell in the basolateral amygdala (BLA) to better understand its role in the reinstatement of cocaine-seeking behavior. Studies have shown that the BLA and nucleus accumbens (NAcc) play a key role in cue-primed reinstatement. Therefore, once the BLA model is complete, a network level model involving the BLA and NAcc will be developed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In 1964, NASA discovered that sufficient cooling of an astronaut could not be accomplished via air convection in the suit due to ventilation. Thus, NASA upgraded to the liquid cooling garment (LCG) to provide better thermal comfort. The astronaut controls the temperature of the water flowing through the LCG by manually adjusting the temperature control valve (TCV) located on the front of the space suit. However, manually manipulating the TCV setting can dramatically reduce the productivity of the astronaut, distracting them from the extra-vehicular activity (EVA) mission at hand. In general, humans are poor judges of their thermal state and since an astronautÃ¢Â?Â?s attention is focused on other things during EVA, there are time lags in the application of temperature control. This can cause overheating or overcooling; placing unneeded thermal stress on the astronaut, and possibly degrading EVA performance. An attractive solution to this problem is the use of automatic control for thermal comfort. Before an automatic controller can be designed, human thermal models are needed to accurately describe the transient heat transfer processes and predict the thermal response within an acceptable band of accuracy. Such human thermal models can also be extended to predict thermal stress for various applications involving fire fighters, combat pilots, hazardous waste workers, and other relevant industries. Hysterical paralysis is a conversion disorder characterized by some lack of motor functioning brought about by psychological mechanisms without the presence of organic injury in the central nervous system. In conversion disorders, it is assumed that the victim when presented with distressing mental conflict, which seem insurmountable, converts this anxiety into a somatic symptom to resolve or alleviate the tension. It is important to note that these physical symptoms are not intentionally produced to malinger. In many instances, the victim is unaware of the stressor that might have caused the symptoms. An assumption can be made that there must be some neurobiological dysfunction that can explain hysterical paralysis. So, mathematical modeling could be the first step to elucidate the neurobiology involved with conversion motor paralysis. In turn, identifying the specific location of brain lesions that produce hysterical paralysis can aid in the diagnosis of this disorder. The rate of return to craving and drug seeking behavior in cocaine addicts is high (Dackis and O'Brien, 2001), and is further exacerbated by exposure to certain environmental stimuli. Three types of environmental stimuli that can precipitate relapse include drug exposure, stress, and drug-associated cues. The modeling of the basolateral amygdala and nucleus accumbens network would benefit the research community as it is implicated in cue-primed reinstatement of cocaine seeking behavior.
Task Progress:	In the human thermal modeling front, a comparison in cold-induced vasoilation response (CIVD) response between predicted and measured finger temperatures was conducted and it was shown that there is a fairly good agreement, indicating that the model is capable of showing CIVD responses in the fingers. As anticipated, the predicted response of finger temperature without CIVD is lower than the finger temperature response with CIVD during the oscillatory stage. The obvious conclusion is that the CIVD mechanism reduces extremity discomfort and prevents injuries due to cold exposure. A study was conducted by the author and colleagues for the purpose of determining tolerance limits for subjects experiencing extreme heat stress. Participants (7 males, 7 females, 19 – 26 years old) exercised (200 and 425 W) at three different environments (normal – 23.9ºC, 45% RH; jungle – 35ºC, 75% RH; and desert – 43.3ºC, 20% RH) while wearing heavy clothing. A neural network predictive model based on the experimental data was developed. In addition, another objective was to determine whether previously developed heat strain models like the Givoni-Goldman prediction can accurately predict tolerance limits in comparison to experimental data obtained by the author and colleagues. Both models predict the response of the 'average' individual, thus correction factors based on individual characteristics (i.e. fitness, body type, acclimation status, etc) were incorporated to improve the prediction of heat strain. In addition, past heat strain predictive models and individual characteristics that effect tolerance limits were extensively reviewed by the author.
Bibliography Type:	Description: (Last Updated: 06/22/2021)
Articles in Peer-reviewed Journals	Iyoho AE, Nair SS, Xu D, Nair J, Beitman B. "Computational modeling of hysterical paralysis." Conf Proc IEEE Eng Med Biol Soc. 2004;2004:894-7. (26th International Conference of the IEEE Engineering in Medicine and Biology Society, San Francisco, CA, September 2004.) https://doi.org/10.1109/IEMBS.2004.1403303 ; PMID: 17271822 , Sep-2004
Papers from Meeting Proceedings	Iyoho AE, Jang TS, Nair SS. "Human thermal model with extremities for asymmetric environments." American Controls Conference, Boston, MA, July 2004. American Controls Conference, July 2004. , Jul-2004